

What we solved, and how

We're hell0, four bachelor's students at Aalto with no quantum background before this weekend. We solved 41 of the 49 circuits, all on our laptops. Here's the full breakdown and the methods behind it.

Team hell0 For QMill · david.karpuk@qmill.com Solved 41 / 49

01 / Results

Every circuit, and what we used

#	CIRCUIT	Q	METHOD	STATUS
VERY EASY				
1	8_1	8	statevector	✓
2	16_2	16	statevector	✓
3	24_3	24	statevector	✓
4	28_4	28	MPS sweep	✓
5	32_5	32	MPS + verify	✓
6	36_6	36	MPS + verify	✓
7	40_7	40	MPS sweep	✓
8	48_8	48	MPS + verify	✓
9	56_9	56	MPS + verify	✓
10	64_10	64	MPS + verify	✓
EASY				
11	8_11	8	statevector	✓
12	16_12	16	statevector	✓
13	24_13	24	statevector	✓

#	CIRCUIT	Q	METHOD	STATUS
14	32_14	32	MPS sweep	✓
15	36_15	36	Pauli backprop	✓
16	40_16	40	MPS sweep	✓
17	40_17	40	MPS sweep	✓
18	40_18	40	MPS marginals	✓
19	48_19	48	MPS marginals	✓
20	48_20	48	MPS + verify	✓
21	48_21	48	MPS marginals	✓
22	56_22	56	MPS + verify	✓
23	56_23	56	MPS marginals	✓
24	56_24	56	MPS marginals	✓
25	64_25	64	permuted MPS	✓
26	64_26	64	permuted MPS	✓
MODERATE				
27	8_27	8	statevector	✓
28	16_28	16	statevector	✓
29	24_29	24	MPS	✓
30	32_30	32	MPS	✓
31	48_31	48	MPS	✓
32	48_32	48	MPS	✓
33	56_33	56	MPS	✓
34	64_34	64	MPS	✓
HARD				
35	40_35	40	Pauli / MPS	✓
36	48_36	48	Pauli / MPS	✓
37	48_37	48	Pauli / MPS	✓
38	56_38 ★	56	Pauli backprop	✓

#	CIRCUIT	Q	METHOD	STATUS
39	56_39	56	Pauli backprop	✓
40	64_40	64	Pauli / MPS	✓
41	64_41	64	Pauli backprop	✓
OPEN — UNSOLVED BY EVERY TEAM				
42	48_42	48	MPO cancellation (unfinished)	open
43	56_43	56	MPO cancellation (unfinished)	open
44	64_44	64	—	open
45	72_45	72	—	open
46	80_46	80	—	open
47	88_47	88	—	open
48	96_48	96	—	open
49	104_49	104	—	open

★ 56_38 we believe no other team cracked.

02 / Methods

The toolkit

We matched a method to each circuit rather than leaning on one, and started every circuit with a cheap structural triage that told us which tool was worth running.

TRIAGE

Odd-parity defect count. Count the qubit pairs touched by an odd number of CX gates — the entangling links the inverse-pair structure can't cancel. It reads off the gate list in milliseconds and predicts a circuit's real difficulty far better than its qubit or gate count, so we never spent compute on the wrong method. `defect_count.py`

WHAT CRACKED THE 41

Exact statevector. Small circuits: build the full state, read the peak directly.

Bond-capped MPS sweeps. Greedy max-marginal decoding through the mid-range; the peak survives truncation because these outputs are so concentrated. `crack.py`

Spectral (Fiedler) MPS reordering. Relabel the qubit line by the CX interaction graph so frequently-coupled qubits sit adjacent, which removes most of the SWAP churn that dominates MPS on dense long-range circuits — this got the 64-qubit ones. `mps_marginal2.py`

$\langle Z_i \rangle$ marginal decoding. Read each qubit's average and threshold it; cheap and parallel, used across several 48–56q circuits.

Find-cheap / verify-exact. A candidate can come from a cheap approximate run, but we certify it with one truncation-free single-amplitude contraction $|\langle s | C | 0 \rangle|^2$ — above 0.5 proves the peak is unique. This certified a 64-qubit answer on a laptop. `crack.py`

Vectorized Pauli backpropagation. Push each qubit's measurement back through the circuit as a sum of Pauli terms (scaled to ~1M terms), read the bit from the sign. Cracked `56_38`, the hardest circuit; with no exact check possible, rising the term cap until the answer stops changing was our trust criterion. `pauli_fast_b.py`

BUILT FOR THE TAIL (42–49, UNFINISHED)

Light-cone bulk cancellation. Tensor-network simplification cancels the even-parity bulk exactly, so most qubits collapse to a cheap exact contraction. `lightcone_z.py`

Two-stage cancellation. Simplify the cancelled state once and reuse it per qubit, instead of re-simplifying for each one. `fast_z2.py`

Defect-graph reduced density matrix. Compute $\langle Z_i \rangle$ from the small simplified defect cone. We also tested belief propagation here; it diverges on these signed-amplitude networks, so we fell back to exact contraction of the cone. `bp_z.py`

Two-sided MPO cancellation + unswapping. Collapse the inverse-pair bulk and recover the hidden permutation (after Kremer–Dupuis). Validated and converging, but ~hours per 20,000-gate circuit, and we ran out of compute. `pcsim/unswap.py`

RULED OUT, AND WHY

No Clifford shortcut (the angles aren't near $\pi/2$ multiples). Qiskit's `03` removes zero two-qubit gates. Commutation-aware cancellation won't telescope through the rewrites. Deleting even-parity CX is confidently wrong. Each dead end told us where the real difficulty sat.

THE GRADER

We reverse-engineered the submission API: it scores by sampled XEB (not the exact match the rules describe), a wrong guess leaks nothing, and the reward is `qubits × 1.5difficulty`. We used it as a yes/no oracle to pin the last uncertain bit of `56_38`.

The two ideas we'd reuse anywhere are the defect count (a quick classical read on whether an obfuscated circuit is genuinely hard) and the find-cheap / verify-exact split

— both work on any peaked circuit, and the even-parity fingerprint belongs to any obfuscation built from inverse pairs and rotations.

Team hell0 · Quantum Hack Challenge · for QMill

Gharibyan et al., *Heuristic Quantum Advantage with Peaked Circuits* (2025), arXiv:2510.25838

Kremer & Dupuis, *Efficient Classical Simulation of Heuristic Peaked Quantum Circuits* (2025), arXiv:2604.21908